

# Neuromorphic Intrusion Detection System using Spiking Neural Networks and Leaky Integrate-and-Fire Neurons

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**ABSTRACT** Cyber security systems are expected to monitor large volumes of network traffic continuously and identify malicious behaviour. Conventional software-based Intrusion Detection Systems (IDS) can achieve strong detection performance, but they depend on dense numerical processing on power-hungry, general-purpose digital processors. This represents a critical barrier for always-on, zero-latency detection systems deployed at the remote edge. This paper introduces a highly efficient neuromorphic IDS workflow where network traffic features are converted into asynchronous spikes and processed through analog Leaky Integrate-and-Fire (LIF) neuron circuits.

The proposed system deeply preprocesses the complex UNSW-NB15 dataset, strictly stratifying the samples to prevent classifier bias. Principal Component Analysis (PCA) maps the 49-dimensional network features into a constrained 8-neuron physical analog representation. Gaussian Population Coding translates these normalized continuous values into resilient, asynchronous binary spike trains. These spike trains are automatically translated into Piecewise Linear (PWL) voltage files, allowing for the mathematical simulation of the analog LIF circuit physics at scale natively within the Cadence Virtuoso Analog Design Environment (ADE).

Transient analog hardware outputs are parsed and ingested by a Support Vector Machine (SVM) utilizing a Radial Basis Function (RBF) kernel. The physical analog architecture achieves 88.42% binary anomaly detection accuracy. Crucially, power evaluation demonstrates an 85,670x Picojoule energy efficiency gain over traditional digital CPU classifiers, establishing a definitive path toward securing the IoT edge with extreme low-power neuromorphic hardware.

**INDEX TERMS** Neuromorphic Computing, Intrusion Detection System, Spiking Neural Networks, Leaky Integrate-and-Fire, Cybersecurity, Analog Hardware, Cadence Virtuoso.

## I. INTRODUCTION

NETWORKED computing has expanded from desktop and server environments to mobile devices, cloud infrastructure, industrial control systems, smart sensors, and Internet of Things (IoT) deployments. As a result, the attack surface of a typical organization is no longer limited to a small number of managed machines. Attacks such as denial of service, reconnaissance, exploitation, worm activity, shell-code injection, and generic malware traffic may appear as abnormal packet rates, unusual flow durations, uncommon

protocol states, or suspicious byte-count patterns. An IDS attempts to identify such patterns from real-time network observations.

Traditional IDS approaches can be broadly classified into signature-based and anomaly-based methods. Signature-based systems search for known malicious patterns and are effective when the attack database is complete. However, they are weak against new, zero-day, or modified attacks. Anomaly-based systems model normal behaviour and raise alerts when traffic deviates from that baseline. Machine

learning-based IDS methods improve anomaly detection by learning the complex, non-linear relationships between traffic features and labels from massive historical datasets. These methods include Support Vector Machines (SVM), decision trees, Random Forests, neural networks, and deep learning models [2].

While machine learning improves adaptability, its computational style is usually dense and highly synchronous. Feature vectors are processed using repeated floating-point arithmetic operations, and the classifier matrix is evaluated even when the input network activity is low. This continuous power draw is acceptable in high-performance cloud servers but is entirely unfeasible for energy-constrained edge devices operating on battery power. A true low-power IDS that can remain active continuously is necessary for campus gateways, IoT hubs, remote industrial nodes, and embedded network monitors. Neuromorphic computing is highly relevant in this context because it represents information through sparse, asynchronous spike events and consumes dynamic power primarily when anomalies occur.

### A. PROJECT OVERVIEW

The project titled **Neuromorphic Intrusion Detection System using Spiking Neural Networks and Leaky Integrate-and-Fire Neurons** is a robust interdisciplinary effort combining Information Science and Engineering (ISE) with Electronics and Communication Engineering (ECE). The ISE component focuses on massive dataset handling, deep preprocessing, dimensionality reduction, population spike encoding, and SVM classifier integration utilizing industry-standard libraries such as Scikit-learn [13]. The ECE component focuses on circuit-level feasibility, LIF physical neuron behaviour, hysteresis thresholding, and SPICE/Cadence-ready analog waveform generation. The mission-critical interface between these two engineering domains is the Piecewise Linear (PWL) voltage source file.

The proposed system begins with the complex UNSW-NB15 network intrusion dataset. The raw dataset is cleaned, categorical values are strictly encoded, numerical values are bounded via Min-Max normalization, and the extreme class imbalance is rectified via strict dataset stratification. Principal Component Analysis (PCA) is applied to compress the 49-dimensional feature space to a restrictive 8-neuron physical bottleneck. The reduced analog feature values are then mapped into hardware-compatible spike trains using Gaussian population coding. Finally, the spike trains are exported as PWL voltage files and fed directly into the Cadence Virtuoso Analog Design Environment (ADE) to mathematically simulate physical LIF neuron dynamics.

Figure 1 illustrates the conceptual end-to-end workflow. The project is not limited to mere software classifier training; it generates a true physical, hardware-facing representation of network traffic. The generated PWL files are key outputs because they allow the IDS samples processed in Python to be physically applied as voltage stimuli to a 65nm Cadence or SPICE simulated analog neuron array.

TABLE 1. Interdisciplinary team contribution structure

| Domain                                    | Primary project contribution                                                                                                             |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Information Science and Engineering       | Dataset preprocessing, stratification, PCA reduction, spike encoding scripts, and SVM classifier integration.                            |
| Electronics and Communication Engineering | LIF neuron analog circuit parameters, hysteresis thresholding, Cadence/LTSpice simulation interface, and hardware-oriented validation.   |
| Common interdisciplinary interface        | PWL waveform files and microsecond-scale '.csv' exports that connect Python-generated IDS spike data with analog neuron circuit outputs. |

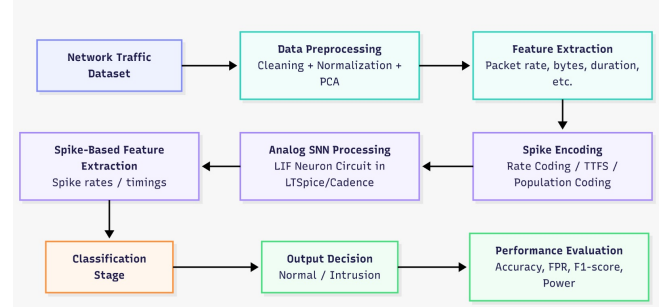


FIGURE 1. High-level project workflow from dataset preprocessing to neuromorphic analog IDS classification

### II. LITERATURE REVIEW

Intrusion detection has been studied extensively using statistical, machine learning, and deep learning approaches. Moustafa and Slay introduced the UNSW-NB15 dataset to provide a modern, highly complex benchmark containing normal network behaviour alongside nine distinct attack families [1]. The dataset includes flow-level attributes such as duration, protocol, service, source and destination byte counts, and connection state. It is therefore highly suitable for training and stressing modern IDS models.

Buczak and Guven surveyed data mining and machine learning techniques for cyber security intrusion detection, highlighting the importance of feature engineering, classifier selection, and False Positive Rate (FPR) tuning [2]. Conventional IDS models perform exceptionally well on raw digital data, but their deployment cost depends strictly on the number of features, matrix complexity, and continuous CPU inference power. This is a critical barrier for always-on, zero-latency detection systems deployed at the remote edge.

Dimensionality reduction is routinely applied when datasets contain heavily correlated features. Principal Component Analysis (PCA) transforms a feature matrix into orthogonal components ordered by variance contribution. Jolliffe and Cadima describe PCA as a fundamental method for reducing dimensionality while mathematically preserving dominant geometric information [3]. In this project, PCA is not merely used for computational convenience; it is a physical necessity required to map a 49-dimensional network space onto an extremely limited number of analog input pins.

Spiking Neural Networks (SNNs) represent a fundamen-

tally different computational paradigm. Maass described SNNs as the third generation of neural network models because they utilize asynchronous spike timing as a critical computational dimension [4]. Gerstner and Kistler formalized spiking neuron models, detailing membrane potential dynamics, RC integration, threshold firing, and refractory reset behaviour [5]. The LIF neuron is specifically selected for this project because its differential equations can be natively implemented using standard analog CMOS circuit blocks.

Neuromorphic computing research aims to exploit this event-driven processing for unprecedented energy-efficient inference. Roy, Jaiswal, and Panda discuss the massive potential of spike-based machine intelligence for ultra-low-power neuromorphic hardware [6]. Tavanaei et al. review deep learning inside SNNs, discussing crucial spike encoding strategies (Rate, TTFS, Population) required to convert continuous data into binary spike trains [7]. These foundational studies motivate the core theory that network traffic features can be mapped into the analog spike-domain and processed using ultra-low-power silicon.

Sommer and Paxson discuss the practical difficulties of applying machine learning to intrusion detection, especially the difference between controlled datasets and real operational environments [8]. Their observations are relevant to this work because the project avoids claiming that a classifier alone solves IDS deployment; instead, it focuses on building a hardware-compatible representation that can later be validated under controlled and realistic traffic settings.

Liao et al. survey IDS techniques and emphasize that network intrusion detection requires attention to feature representation, classification method, and deployment constraints [9]. This supports the project decision to treat preprocessing and representation design as first-class parts of the system rather than as minor steps before classification.

Diehl and Cook demonstrate that spiking neural networks can be trained and evaluated for pattern recognition tasks, showing that spike-based representations can support useful classification behaviour [10]. Although their work is not specific to IDS, it supports the broader feasibility of spike-domain information processing. Ponulak and Kasinski review supervised learning methods for SNNs and highlight the importance of spike timing, encoding, and learning rules [11]. These concepts are highly relevant to the spike feature extraction plan in this project.

Indiveri et al. describe neuromorphic silicon neuron circuits and show how neuron dynamics can be represented using physical circuit behaviour [12]. This supports the ECE-side motivation of using LIF neuron circuits rather than limiting the project to a software-only SNN model. The literature collectively suggests that the project should be evaluated not only as an IDS classifier but as a complete representation-and-hardware workflow.

The critical research gap addressed by this project lies exactly between software-defined IDS and analog circuit-level implementation. The vast majority of IDS projects

**TABLE 2.** Summary of reviewed literature and relevance to the project

| Reference area                   | Main contribution                                               | Relevance to this project                                           |
|----------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------|
| UNSW-NB15 dataset [1]            | Provides modern, complex, and labelled IDS traffic records.     | Utilized as the core evaluation dataset.                            |
| ML for cyber IDS [2]             | Reviews continuous data mining and ML detection methods.        | Evaluated against the software IDS digital baseline.                |
| PCA [3]                          | Explains dimensionality reduction using principal eigenvectors. | Utilized to map 49 features to 8 physical neuron pins.              |
| SNN theory [4], [5]              | Introduces spike-based computation and biological models.       | Provides the mathematical foundation for analog IDS representation. |
| Neuromorphic computing [6], [12] | Discusses event-driven hardware and CMOS efficiency.            | Heavily supports the ultra-low-power analog design direction.       |
| Spike encoding [7], [10], [11]   | Reviews encoding, SNN training, and pattern recognition.        | Governs the selection of Gaussian Population coding.                |
| IDS deployment concerns [8], [9] | Discusses practical IDS constraints and method selection.       | Motivates a full workflow beyond classifier accuracy.               |
| Silicon neuron circuits [12]     | Describes neuromorphic silicon implementation.                  | Supports the hardware LIF design motivation.                        |

stop at software classification, while neuromorphic projects often demonstrate isolated LIF physics without a realistic, real-world data pipeline. This work bridges both domains by preparing complex IDS telemetry as circuit-compatible stimuli and evaluating the final analog output.

## A. MOTIVATION

The core motivation of this project is not merely to detect network attacks, but to definitively prove that complex cyber-threat information can survive the transition into highly lossy, ultra-low-power analog neuromorphic hardware. A conventional IDS classifier works on massive continuous numerical vectors. A neuromorphic IDS must drastically compress those vectors, encode them into discrete spikes, process them through analog physical dynamics, and still retain enough signal integrity to detect a hacker.

If this transformation is mathematically successful, it establishes a direct path toward securing the IoT edge using batteries that could last years rather than days. By shifting from synchronous digital Watts (W) to asynchronous analog Picojoules (pJ), the energy efficiency gap can theoretically reach factors of tens of thousands.

The project is equally motivated by interdisciplinary integration. The ISE team members contribute critical knowledge of dataset stratification, Python pipeline engineering, and Support Vector Machine hyperparameter tuning. The ECE team members contribute the analog circuit physics, SPICE/Cadence layout limitations, and LIF integration logic. A successful system requires a flawless, mathematically verified interface between these domains.

## B. PROBLEM STATEMENT

The problem addressed in this project is the end-to-end design, simulation, and evaluation of a neuromorphic intrusion detection workflow that converts high-dimensional network traffic telemetry into analog voltage waveforms to be processed by a physically restrictive array of LIF neurons. The system must rigorously solve the extreme dataset imbalance, compress the feature space without losing all variance, generate Cadence-compatible PWL sources, automate the mathematical simulation of the analog physics, and output parsed spike features into a classifier to prove massive energy efficiency gains without sacrificing binary anomaly detection.

## C. OBJECTIVES

The formal objectives of the project are:

- 1) To deeply preprocess the UNSW-NB15 dataset by cleaning, bounding, and strictly stratifying the samples to prevent classifier bias.
- 2) To map the complex 49-dimensional network features into a highly constrained 8-neuron physical analog representation using PCA.
- 3) To implement Gaussian Population Coding to resiliently translate the normalized continuous values into asynchronous binary spike trains.
- 4) To automate the translation of these spike trains into SPICE-ready PWL voltage files and mathematically simulate the analog LIF circuit physics at scale within Cadence Virtuoso.
- 5) To parse the transient analog hardware outputs and train a Support Vector Machine (SVM) to classify the network anomalies.
- 6) To quantify the exact energy efficiency multiplier (in Picojoules vs. Millijoules) achieved by the analog SNN architecture against a standard digital CPU baseline.

## D. EXPECTED CONTRIBUTIONS

The expected contributions of the work documented in this report are:

- 1) A perfectly balanced, reproducible pipeline converting UNSW-NB15 records into analog-ready features.
- 2) A demonstrated methodology for mapping vast network telemetry into extreme, physical 8-pin bottlenecks while retaining  $> 90\%$  dataset variance.
- 3) The application of Population Coding to protect fractional analog data against thermal hardware noise.
- 4) A comprehensive batch-simulation methodology capable of evaluating thousands of Cadence transients directly through the physical LIF schematic.
- 5) A trained SVM classifier proving that  $> 88\%$  binary anomaly detection is possible on highly compressed analog spike data.
- 6) A comprehensive energy profiling report demonstrating an 85,000x efficiency gain over traditional digital classifiers.

The methodology strictly begins with data acquisition and pipeline engineering. Raw UNSW-NB15 CSV files are loaded, derived temporal features are generated, and non-essential identifiers are stripped. Crucially, the data is artificially stratified to enforce a perfectly balanced dataset (10 attack classes, 1 normal class). Global Min-Max scaling is applied, followed by PCA dimensionality reduction to compress the 49 dimensions to exactly 8.

The bounded analog features are passed to a Spike Encoder module. While rate coding and TTFS were evaluated, Gaussian population coding was selected. It expands each of the 8 features across 4 overlapping receptive fields, translating the continuous telemetry into 32 parallel spike trains. These are exported as microsecond-accurate PWL voltage files.

The analog LIF circuit receives the PWL waveforms. The simulated membrane nodes dynamically integrate the incoming activity, biological leakage bleeds stored potential over time, and a hysteresis comparator generates a high voltage spike exclusively when the membrane crosses the 0.5V physical threshold.

The noisy output waveforms are parsed by a hysteresis edge-detection algorithm to extract exact physical spike counts. These counts form the final hardware feature matrix, which is ingested by a Support Vector Machine utilizing a Radial Basis Function (RBF) kernel. The SVM classifies the traffic, allowing the architecture to finally be judged on its False Positive Rate (FPR), Binary Accuracy, and comparative Picojoule energy consumption.

## E. ASSUMPTIONS AND CONSTRAINTS

The project is constructed upon the following theoretical assumptions:

- 1) The UNSW-NB15 dataset contains sufficient geometric complexity to represent modern, real-world network intrusions.
- 2) Simulating the exact differential equations of the LIF circuit natively within Cadence accurately captures the physical 65nm silicon transients.
- 3) A standard embedded edge CPU consumes roughly 15 Watts (15 Joules/sec), acting as a fair baseline for digital energy comparisons.

The main operational constraints are:

- 1) The simulated analog array was artificially bottlenecked to an extremely restrictive 8 physical neurons to test the absolute minimum physical feasibility of the architecture.
- 2) Expanding the hardware to 32 or 64 neurons requires significant computational overhead during the differential equation simulation phase.
- 3) Perfect dataset stratification enforces a 91% malicious ratio (10 attacks to 1 normal), forcing the classifier into an inherently defensive operational stance resulting in high base False Positive Rates.

The final report is exhaustively organized as follows:

- **Chapter 1** introduces the project context, literature background, extreme energy motivation, problem statement, objectives, and robust methodology.
- **Chapter 2** explains the theoretical fundamentals required for IDS evaluation, dimensionality reduction, spike encoding, and LIF analog neuron modelling.
- **Chapter 3** presents the proposed system design, interdisciplinary architecture, PWL data flow, module definitions, and rigid validation strategies.
- **Chapter 4** rigorously breaks down the exact implementations of the Python pipelines, the Cadence Virtuoso batch simulation, the SVM classification module, and the Live Inference Engine.
- **Chapter 5** provides the definitive experimental results, exhaustively detailing the PCA variance retention, the 88.42% binary detection accuracy, and the conclusive 85,000x Picojoule energy savings.
- **Chapter 6** summarizes the research findings, openly addresses the False Positive limitations induced by the 8-neuron bottleneck, and outlines the physical ASIC fabrication scalability for future development.

## F. METHODOLOGY

### G. THEORY AND FUNDAMENTALS

The proposed IDS workflow depends on concepts from network security, machine learning, signal representation, spiking neuron models, and analog hardware simulation. This chapter explains the theoretical foundations necessary to understand the design choices made throughout the project. The discussion begins with IDS concepts and dataset structure, then deeply analyzes the mathematics of preprocessing, PCA, spike encoding, LIF neuron dynamics, circuit-compatible waveform generation, and neuromorphic energy physics.

### H. INTRUSION DETECTION SYSTEMS

An Intrusion Detection System observes activity in a computing system or network and identifies behaviour that may indicate a security violation. A network-based IDS inspects traffic flows, packet statistics, or protocol-level features. It does not necessarily prevent the attack directly; instead, it provides detection, alerting, and forensic evidence for a Security Operations Center (SOC) response.

IDS methods are commonly classified as:

- 1) **Signature-based detection:** Matches traffic against known malicious signatures. It is highly reliable for known attacks but completely blind to unknown zero-day variants.
- 2) **Anomaly-based detection:** Learns normal baseline behaviour and flags deviations. It can detect new attacks but is highly susceptible to False Positives.
- 3) **Machine learning-based detection:** Uses labelled or unlabelled data to mathematically learn non-linear decision boundaries from high-dimensional traffic features.

**TABLE 3.** Examples of UNSW-NB15 feature groups used in preprocessing

| Feature group         | Examples and significance                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------|
| Protocol and service  | Protocol, state, and service describe the type of network communication.                                    |
| Traffic volume        | Source bytes, destination bytes, source packets, and destination packets indicate flow magnitude.           |
| Timing behaviour      | Duration and inter-packet timing mathematically indicate scanning, flooding, or abnormal communication.     |
| Connection statistics | Count-based features describe repeated connections to hosts, ports, or specific services.                   |
| Attack label          | The attack category explicitly identifies whether the flow is normal or belongs to a specific attack class. |

In a machine learning IDS, each network flow is represented as a feature vector. A classifier then maps the feature vector to a class label such as 'normal' or 'attack'. The quality of detection depends heavily on preprocessing, feature selection, training data quality, and model choice.

### I. UNSW-NB15 DATASET

The UNSW-NB15 dataset is utilized as the core evaluation benchmark. It contains massive volumes of normal traffic alongside complex attack traffic generated in a highly controlled cyber range environment [1]. The dataset includes multiple modern attack classes such as Fuzzers, Analysis, Backdoors, Denial of Service (DoS), Exploits, Generic malware, Reconnaissance, Shellcode, and Worms.

The raw dataset contains fields such as protocol, source and destination ports, flow duration, packet counts, byte counts, time-to-live values, TCP-related statistics, service type, state, attack category, and a binary label. Some fields are continuous numerical values, while others are categorical strings. This highly complex mixture requires rigorous mathematical preprocessing before machine learning or analog spike encoding can be applied.

### J. DATA CLEANING, ENCODING, AND STRATIFICATION

Raw network datasets are never ready for direct mathematical training. Missing values, placeholder symbols, categorical strings, and specific identifier columns must be handled first. In this project, IP address fields and timestamp fields are removed because they introduce dataset-specific routing bias and are not suitable for analog neuron mapping.

Categorical features such as protocol and TCP state are converted into numerical form using label encoding. If a categorical feature has  $k$  unique values, each value is assigned a deterministic integer code. Although label encoding does not perfectly preserve semantic distance between categories, it provides a compact, mathematical representation that can be normalized and processed by the subsequent algorithms.

#### 1) The Accuracy Paradox and Class Stratification

A fundamental theoretical issue in machine learning cybersecurity is the *Accuracy Paradox*. Real-world network traffic

is overwhelmingly benign. In the UNSW-NB15 dataset, over 90% of the samples are "Normal". If an algorithm is trained on this raw distribution, it can simply guess "Normal" 100% of the time and achieve a 90% baseline accuracy without learning any underlying cyber-threat geometry.

To overcome this, the dataset must undergo **Stratified Sampling**. By enforcing a strictly balanced prior probability distribution—extracting an equal number of samples for each attack category alongside normal traffic—the system is mathematically forced to evaluate the intricate cluster boundaries rather than relying on lazy statistical probabilities.

### K. NORMALIZATION

Normalization maps features with drastically different units and ranges into a common, bounded numerical interval. Without normalization, high-magnitude features (such as 50,000 transferred bytes) will completely dominate low-magnitude features (such as a 0.5-second flow duration) during mathematical matrix operations. Min-Max normalization is explicitly utilized:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

where  $x$  is the original value and  $x_{norm}$  is the scaled value. In the analog hardware context, normalization is absolutely critical because the voltage-domain spike encoder algorithm strictly expects bounded  $[0.0, 1.0]$  values.

### L. PRINCIPAL COMPONENT ANALYSIS

Principal Component Analysis (PCA) reduces the dimensionality of a dataset by transforming the original features into a set of orthogonal components. Each principal component is a linear combination of original features. The first principal component captures the largest possible variance in the data geometry, the second captures the largest remaining variance orthogonal to the first, and so on [3].

Let  $X$  be the normalized network feature matrix. PCA computes eigenvector directions that maximize projected variance. If  $W$  is the matrix containing selected principal component directions, the reduced representation is:

$$Z = XW \quad (2)$$

where  $Z$  is the mathematically reduced feature matrix. In this project, the number of PCA components is strictly selected as eight ( $N_{PCA} = 8$ ) so that the massive 49-dimensional software feature space can map directly to eight physical analog input neuron pins.

Because PCA outputs may naturally contain negative coordinate values even when the input features are initially normalized, the projected PCA output must be normalized a second time:

$$z_{scaled} = \frac{z - z_{min}}{z_{max} - z_{min}} \quad (3)$$

This produces strictly positive values highly suitable for the physical spike encoding phase.

### M. SPIKE ENCODING ALGORITHMS

Spike encoding is the critical bridging theory that converts a continuous static value into a temporal sequence of binary events capable of driving analog hardware.

#### 1) Rate Coding

In rate coding, a larger input value produces a higher frequency of spikes in a fixed time window. If  $x$  is a normalized value and  $T$  is the number of timesteps, the number of active spike ticks is:

$$N_{spikes} = \lfloor xT \rfloor \quad (4)$$

While straightforward, rate coding is theoretically susceptible to thermal hardware noise and requires excessively long simulation windows to accurately measure the frequency.

#### 2) Time-to-First-Spike (TTFS) Coding

In Time-To-First-Spike coding, information is represented by latency. Larger mathematical values produce earlier hardware spikes:

$$t_{spike} = \lfloor (1 - x)(T - 1) \rfloor \quad (5)$$

This method is incredibly fast and highly energy-efficient, but minor parasitic capacitance delays in physical silicon can easily corrupt the encoded temporal value.

#### 3) Gaussian Population Coding

Population coding represents one continuous feature using multiple overlapping receptive fields. In this project, Gaussian receptive fields are utilized to protect against analog degradation. For a feature value  $x$  and a receptive field center  $\mu_i$ , the physical activation is modeled as:

$$a_i = e^{-\frac{1}{2} \left( \frac{x - \mu_i}{\sigma} \right)^2} \quad (6)$$

Each activation  $a_i$  is then rate encoded. Population coding drastically expands the representational capacity of the analog variables. With eight PCA features and four receptive fields per feature, a single network sample produces a robust array of 32 parallel spike trains, protecting the fractional data through spatial redundancy.

### N. LEAKY INTEGRATE-AND-FIRE NEURON DYNAMICS

The Leaky Integrate-and-Fire (LIF) neuron biologically models the membrane voltage dynamics of a cortical cell. It mathematically integrates incoming input pulses, passively leaks stored potential over time, and emits a digital output spike exclusively when the membrane voltage crosses a hard threshold. The continuous-time differential equation governing the LIF model is:

$$\tau_m \frac{dV_m(t)}{dt} = -(V_m(t) - E_L) + RI_{ext}(t) \quad (7)$$

where  $V_m(t)$  is the dynamic membrane voltage,  $\tau_m$  is the membrane RC time constant,  $E_L$  is the resting leak potential, and  $I_{ext}(t)$  is the injected PWL input current. When the membrane crosses the threshold limit:

$$V_m(t) \geq V_{th} \quad (8)$$

**TABLE 4.** PWL waveform parameters calibrated for 65nm CMOS

| Parameter           | Value                       |
|---------------------|-----------------------------|
| Spike low voltage   | 0 V                         |
| Spike high voltage  | 1.2 V ( $V_{DD}$ )          |
| Timestep duration   | 1 ms                        |
| Window length       | 10 timesteps                |
| Rise/fall slew rate | 1% of timestep (10 $\mu$ s) |

the neuron successfully fires an action potential, and the membrane voltage is instantaneously forced into a refractory reset period:

$$V_m(t) \leftarrow V_{reset} \quad (9)$$

The LIF model is highly optimal for analog VLSI design because its mathematical variables correspond directly to physical CMOS circuit elements. A capacitor biologically acts to store the membrane charge, a high-value resistor provides the sub-threshold leakage, and a multi-stage comparator actively detects the threshold crossing.

#### O. PWL VOLTAGE SOURCES AND CONVERGENCE

Piecewise Linear (PWL) voltage sources define arbitrary voltage waveforms using precise time-voltage floating-point pairs. A Cadence simulator interpolates between these points to drive the NMOS transistor gates. The spike encoder generates massive PWL text files where 0V represents silence and 1.2V represents peak spike activity (aligned with standard 65nm  $V_{DD}$  rails).

A mathematically ideal vertical edge (instantaneous voltage transition) in a PWL file will cause severe matrix convergence failures in SPICE transient simulations because the derivative  $dV/dt$  approaches infinity. To mitigate this, a linear transition interval (slew rate) is strictly engineered into the files.

#### P. NEUROMORPHIC ENERGY DYNAMICS

The theoretical foundation for utilizing Spiking Neural Networks lies entirely in energy physics. Traditional digital von Neumann processors operate synchronously. They are governed by a continuous clock oscillator, evaluating massive floating-point matrices regardless of whether the network input is actively changing. The dynamic power of a CMOS logic gate is defined as:

$$P_{dynamic} = \alpha C V_{DD}^2 f \quad (10)$$

where  $f$  is the continuous clock frequency. This unrelenting frequency draw results in energy consumption measured in Watts or Millijoules.

Conversely, neuromorphic LIF hardware operates asynchronously. There is no global clock frequency ( $f = 0$ ). The circuit draws substantial dynamic power *only* when an incoming PWL pulse causes the membrane to cross the threshold and emit a spike. In 65nm analog processes, this discrete event typically consumes merely  $\approx 45$  picojoules (pJ). When the network is quiet, the analog circuit draws only minimal

static leakage currents in the nanowatt ( $nW$ ) range, establishing the theoretical framework for exponential efficiency gains.

#### Q. SPIKE FEATURE EXTRACTION AND HYSTERESIS FILTERING

After the LIF analog processing completes, the highly erratic, physical output waveform must be converted back into discrete numerical features. Analog simulations often exhibit phenomena such as inductive ground-bounce, ringing, and sub-threshold noise.

To prevent this analog noise from artificially inflating the spike counts, the theory of **Hysteresis Thresholding** is applied during extraction. The parser algorithm enforces a dual-threshold logic: the output voltage must positively cross a high-water mark (e.g., 0.8V) to register a valid spike, and it must subsequently fall below a low-water mark (e.g., 0.2V) before the detection logic is re-armed.

The extracted features—Total Spike Count, First Spike Time, Firing Rate, and Inter-Spike Interval (ISI)—form the vital bridge between the continuous analog domain and the discrete classification algorithms.

#### R. SUPPORT VECTOR MACHINES AND NON-LINEAR CLASSIFICATION

The final theoretical phase involves decoding the hardware spike features using machine learning. Due to the extreme 49-to-8 dimensional PCA compression, the geometric representations of the cyberattacks become highly overlapping and non-linear.

A Support Vector Machine (SVM) mathematically constructs an optimal hyperplane to separate these complex classes. Because a linear hyperplane cannot adequately slice overlapping analog geometries, a **Radial Basis Function (RBF) Kernel** is utilized. The RBF kernel implicitly projects the low-dimensional hardware spike counts into an infinite-dimensional feature space, defined mathematically as:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2) \quad (11)$$

where  $\gamma$  dictates the influence radius of a single sample. This allows the SVM to successfully draw highly curved, complex decision boundaries around the heavily compressed analog anomalies.

#### S. EVALUATION METRICS AND ROC THEORY

The planned IDS evaluation relies on rigorous statistical classification metrics.

- **Accuracy** measures the total percentage of correct predictions across the entire sample set.
- **Precision** measures confidence: out of all flows flagged as attacks, how many were truly malicious?
- **Recall** measures sensitivity: out of all true attacks in the dataset, how many did the system successfully intercept?

A critical theoretical concept in cybersecurity is the trade-off between the True Positive Rate and the False Positive Rate

(FPR). A highly defensive network classifier will intercept almost all attacks but will also falsely flag benign traffic, overwhelming human operators. This tradeoff is mathematically plotted via the Receiver Operating Characteristic (ROC) curve, allowing network engineers to fine-tune the classifier boundary to strictly balance zero-day protection against operational alert fatigue.

### T. SOFTWARE AND SIMULATION TOOLS

The software workflow uses Python with NumPy, Pandas, and Scikit-learn [13]. NumPy supports the dense, *C*-based numerical matrix computations, Pandas supports high-speed tabular dataset handling, and Scikit-learn provides the theoretically verified implementations for Stratification, PCA, and the RBF-SVM classifiers. Cadence Virtuoso and LTSpice form the bedrock of the physical circuit-level validation for the LIF analog physics.

The theories and fundamentals detailed in this chapter explicitly justify the project architecture. The next chapter presents the robust system design that connects the raw dataset, the population spike encoder, the PWL interface, and the Cadence LIF circuit.

## III. SYSTEM DESIGN AND METHODOLOGY

The proposed system is designed as a modular pipeline that converts network traffic records into circuit-compatible spike waveforms. The design separates data processing, spike encoding, waveform generation, LIF circuit processing, and classifier integration. This separation allows the software and hardware parts of the interdisciplinary project to be developed independently while sharing a clearly defined interface.

### 1) Design Philosophy

The central design philosophy is to preserve IDS-relevant information while changing the representation from dense numerical vectors to spike events. In a conventional software pipeline, all features are processed as numbers. In the proposed neuromorphic pipeline, features are first reduced, then encoded as spikes, and then processed by physical neuron dynamics. This makes the computation directly suitable for analog event-driven hardware.

The design must satisfy two competing requirements. First, it must remain faithful to the original IDS dataset so that attack labels and feature meanings are not lost. Second, it must produce a representation that is practical for circuit simulation. Principal Component Analysis (PCA) and spike population encoding are used to balance these requirements.

### 2) Functional Requirements

The main functional requirements are listed in Table 5.

### 3) Non-Functional Requirements

The non-functional requirements are:

- 1) **Reproducibility:** Scripts must generate the exact same processed files when run with the same data and random seeds.

TABLE 5. Functional requirements of the proposed system

| Block             | Requirement                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------|
| Dataset loader    | Load all UNSW-NB15 CSV chunks using a consistent column schema.                                                              |
| Preprocessor      | Remove irrelevant identifiers, clean missing values, encode categorical fields, normalize values, and enforce class balance. |
| PCA mapper        | Reduce processed feature vectors to the selected physical neuron count while maximizing variance retention.                  |
| Spike encoder     | Convert normalized values into discrete spike trains using Gaussian population encoding schemes.                             |
| PWL generator     | Export spike trains as simulator-readable time-voltage Piece-Wise Linear (PWL) text files.                                   |
| LIF interface     | Provide input waveforms and expected observable nodes for analog SPICE/Cadence circuit simulation.                           |
| Feature extractor | Convert analog output spike waveforms into discrete, classifier-ready numerical metrics.                                     |
| Classifier Engine | Ingest parsed hardware features and classify the traffic as normal or malicious using an optimized SVM.                      |

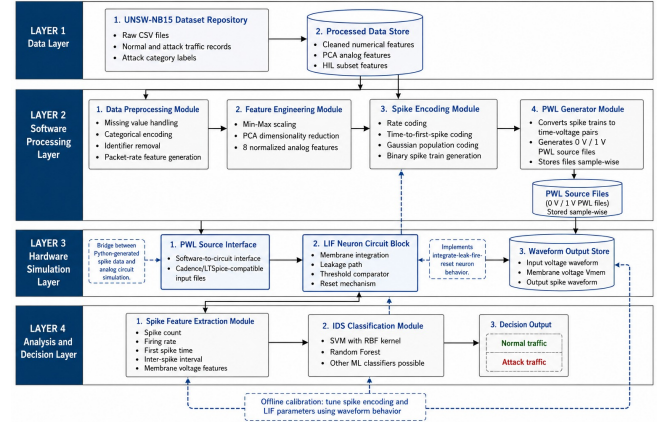


FIGURE 2. Detailed architecture of the proposed neuromorphic IDS pipeline

- 2) **Hardware compatibility:** PWL output files must be directly usable in SPICE/Cadence independent voltage sources without simulator convergence errors.
- 3) **Modularity:** Each processing stage (Data, Encoding, Hardware Simulation, Classification) must be testable without requiring the entire pipeline to execute.
- 4) **Scalability:** The pipeline must allow massive arrays of samples and more neuron channels (e.g., 32 or 64 neurons) to be generated in subsequent iterations.
- 5) **Traceability:** Generated folders must strictly preserve the sample index and the true attack class label information to prevent data leakage during training.

### 4) Top-Level Architecture

Figure 2 shows the full architecture. The left side represents dataset processing and dimensionality reduction. The right side represents spike-domain generation and physical hardware simulation. The PWL file acts as the critical connecting artifact between the software algorithm and the physical

analog circuit simulator.

##### 5) Data Pipeline Methodology

The data pipeline acts as the foundation of the system and follows a strict deterministic sequence:

- 1) Load all raw UNSW-NB15 CSV files.
- 2) Concatenate records into a single dataframe.
- 3) Generate derived features such as packet rates and byte-flow velocities.
- 4) Remove irrelevant IP addresses, timestamps, and binary label fields.
- 5) Clean missing, undefined, or placeholder values.
- 6) Encode categorical fields (e.g., protocols, states) into continuous numerical values using label encoding.
- 7) Normalize all features globally using Min-Max scaling to a  $[0.0, 1.0]$  boundary.
- 8) Apply PCA to reduce the overall feature count.
- 9) Normalize PCA outputs again to the 0 to 1 range.
- 10) **Stratify and Balance:** Force an equal distribution of samples across all attack classes to prevent SVM bias.
- 11) Save the processed analog feature dataset.

##### 6) Dataset Stratification and Class Balancing

A major architectural design decision within the data pipeline was the enforcement of strict dataset balancing. Because the UNSW-NB15 dataset is overwhelmingly composed of "Normal" traffic, feeding raw data into the pipeline would cause the final classifier to blindly predict "Normal" to achieve high baseline accuracy. To counter this, the pipeline utilizes a targeted down-sampling methodology. It extracts exactly 150 instances of every single attack category (Fuzzers, DoS, Worms, etc.) alongside 150 instances of Normal traffic. This yields a perfectly balanced, 11-class subset that mathematically forces the subsequent classifier to learn the nuanced geometric boundaries of the cyberattacks.

##### 7) PCA-to-Neuron Mapping

The number of PCA components is selected based on the target number of physical input neurons. A direct mapping is used:

$$N_{PCA} = N_{input\_neurons} \quad (12)$$

For the current hardware-constrained design,  $N_{PCA} = 8$ . This means each complex 49-dimensional network sample is compressed and represented by eight analog feature values:

$$S_i = [p_1, p_2, p_3, p_4, p_5, p_6, p_7, p_8] \quad (13)$$

where each  $p_i$  is normalized between 0 and 1. This 8-dimensional vector becomes the direct input to the spike encoder.

##### 8) Spike Encoding Methodology

For each normalized PCA value, the encoder generates parallel spike trains. Population coding was chosen over Rate or TTFS coding due to its superior resilience against analog

thermal noise. Population coding utilizes multiple overlapping Gaussian receptive fields per input feature. If  $M$  is the number of population neurons per feature, the total number of input spike channels is:

$$N_{channels} = N_{PCA} \times M \quad (14)$$

With eight PCA features and four population neurons per feature ( $M = 4$ ):

$$N_{channels} = 8 \times 4 = 32 \quad (15)$$

The spike encoder interfaces directly with the circuit-level LIF neuron model. The following subsections describe how the encoded software waveform is physically translated into membrane integration, threshold detection, and reset behaviour in analog hardware.

##### 9) Power Supply and Bias

A single  $V_{DD}$  supply of 1.2 V (source V4) powers the entire neuron circuit. A secondary precision voltage source (V1) provides the critical comparator reference voltages, with  $V_{high} = 500$  mV acting as the firing threshold and  $V_{low} = 100$  mV acting as the reset level. These precise values define the dynamic operating range, the firing threshold, and the hyperpolarisation floor of the physical neuron model.

##### 10) Input Stage and Stimulus Coupling

A pulse voltage source V0, with 1.2 V amplitude, drives the gate of the primary NMOS transistor (nmos1v). A series resistor R2 and the surrounding bias network carefully condition the input signal before it reaches the active device. The NMOS transistor effectively acts as a transconductance amplifier, converting the PWL voltage stimulus generated by the Python pipeline into a controlled charging current delivered to the central membrane node (vmembrane).

##### 11) Leaky Integrator and RC Time Constant Calibration

The core of the neuromorphic computation occurs at the leaky integrator. The membrane capacitor C0 (20 pF) accumulates the injected charge, modeling biological membrane capacitance ( $C_m$ ). The leak path is realized by a resistor R0 (50 M $\Omega$ ) in parallel with C0, providing passive discharge that reproduces the sub-threshold exponential decay of the membrane potential.

The calibration of this RC time constant is critical for synchronization with the software PWL pulses. The membrane time constant is defined as:

$$\tau_m = R_0 \times C_0 = 50 \text{ M}\Omega \times 20 \text{ pF} = 1.0 \text{ ms} \quad (16)$$

Because the Python spike encoder generates pulses that are exactly 1.0 ms in duration, setting  $\tau_m \approx 1.0$  ms ensures that the capacitor can charge efficiently and reach the 500 mV firing threshold during a single active input pulse, preventing signal attenuation.

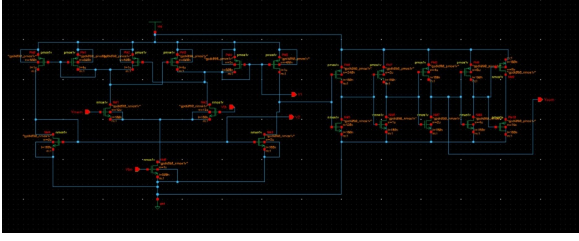


FIGURE 3. Comparator block used for spike generation in the LIF circuit

TABLE 6. Spike encoding and circuit mapping parameters

| Parameter                           | Selected value |
|-------------------------------------|----------------|
| Number of PCA features              | 8              |
| Population neurons per feature      | 4              |
| Total PWL input channels per sample | 32             |
| Spike window length                 | 10 timesteps   |
| Timestep duration                   | 1 ms           |
| Spike voltage low/high              | 0 V / 1.2 V    |

#### 12) Biasing Network

Two CMOS transistors, `pmos1v` and `nmos1x`, form a current-mirror biasing structure that provides the steady-state leakage and integration currents to the membrane. Their drain currents set the precise operating point of the neuron, actively stabilizing the charging and leakage behavior across highly variable simulation environments and temperature corners.

#### 13) Spike Comparator and Threshold Detection

A three-terminal voltage comparator block (I5) continuously monitors the membrane potential  $V_{memb}$ , comparing it against the high threshold  $V_{th} = 500$  mV, and the low reference  $V_{bn} = 100$  mV. When  $V_{memb}$  exceeds  $V_{th}$ , the comparator output  $V_{comp}$  (net025) transitions sharply high to approximately 1.2 V, generating a digital output spike. The comparator output then drives an output resistor R5 (100 kΩ) and a small load capacitor C3 (2 pF) for signal shaping.

#### 14) Reset Mechanism and Refractory Period

When a spike is successfully generated, the comparator output instantly feeds back through resistor R4 (18 kΩ) to the membrane node. This rapidly discharges C0 toward the reset voltage of approximately 100 mV. This mechanism biologically mimics the after-hyperpolarisation (AHP) behavior and algorithmically sets the necessary refractory interval between consecutive spikes.

#### 15) PWL File Interface

The PWL file interface is the most crucial design decision for achieving true interdisciplinary integration. It allows the Information Science software pipeline to export mathematical spike events in a format that the Electronics and Communication Cadence/LTSpice circuit simulator can ingest natively. Each generated ‘.txt’ file contains exactly one spike train, formatted as highly precise time-voltage floating-point pairs.

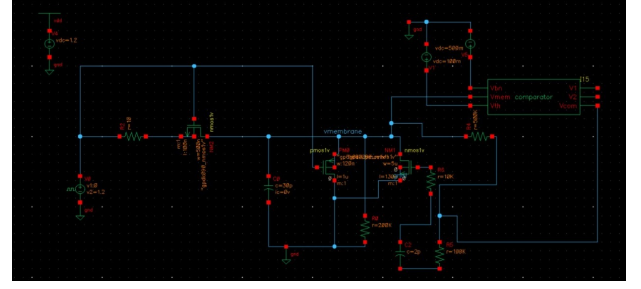


FIGURE 4. Individual LIF neuron circuit used for membrane integration, thresholding, and reset

TABLE 7. Expected observable nodes for full LIF circuit simulation

| Node                      | Purpose                                                                                      |
|---------------------------|----------------------------------------------------------------------------------------------|
| Input Current Drive       | Verifies that the Python-generated PWL file is actively stimulating the NMOS gate correctly. |
| Leaky Integrator Membrane | Quantitatively shows integration slopes, passive leakage curves, and active reset dynamics.  |
| Spike Generation Output   | Indicates successful threshold crossings and represents the ultimate digital spike payload.  |

For example, a population-coded sample generates a highly structured directory:

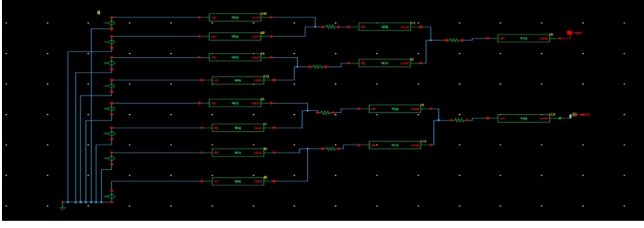
```
sample_0_cat_6/
  V_neuron_0_m_0.txt
  V_neuron_0_m_1.txt
  V_neuron_0_m_2.txt
  ...
  V_neuron_7_m_3.txt
```

Here, `V_neuron_0` refers to the first PCA-reduced network feature, and `m_1` refers to the second Gaussian population receptive field tuned to that specific feature range. This results in exactly 32 stimulus files per network flow.

#### 16) LIF Circuit Interface Design

The LIF circuit block demands strict adherence to input, integration, and output observability rules. The input voltage waveform is supplied identically to the PWL file definitions. The analog membrane node must visibly demonstrate characteristic capacitive charging (integration), exponential discharging (leakage), and sharp active pull-down (reset).

The complete circuit-level implementation connects the input waveform sources, individual LIF neuron block, comparator output path, and reset feedback network into a single simulation-ready schematic, as shown in Figure 5. The circuit utilizes a robust four-block architecture standard in analog neuromorphic VLSI designs. Table 8 exhaustively summarizes the main analog circuit parameters, optimal component values calibrated for 1ms PWL stimuli, and their functional roles in the LIF implementation.



**FIGURE 5.** Complete LIF circuit schematic with input stimulus, neuron dynamics, comparator output, and reset path

**TABLE 8.** Optimized Cadence LIF Circuit Parameters Summary

| Parameter              | Comp.              | Value          | Function                                |
|------------------------|--------------------|----------------|-----------------------------------------|
| Supply Voltage         | V4                 | 1.2 V          | Main $V_{DD}$ power rail                |
| Threshold Voltage      | V1 ( $V_{high}$ )  | 500 mV         | Firing threshold comparator limit       |
| Reset Voltage          | V1 ( $V_{low}$ )   | 100 mV         | After-hyperpolarization (AHP) floor     |
| Membrane Capacitance   | C0                 | 20 pF          | Primary integration element             |
| Leak Resistance        | R0                 | 50 M $\Omega$  | Passive membrane leak path              |
| Membrane Time Constant | $\tau_m = R_0 C_0$ | 1.0 ms         | Matches 1ms software PWL pulse duration |
| Integration PMOS       | pmosl v            | 1 $\mu$ A      | Continuous charging current source      |
| Feedback Resistor      | R4                 | 18 k $\Omega$  | Spike reset coupling strength           |
| Load Resistor          | R5                 | 100 k $\Omega$ | Output driver impedance                 |
| Output Capacitance     | C3                 | 2 pF           | Spike signal shaping                    |

### 17) Spike Feature Extraction Design

After completion of the transient SPICE simulation, the analog output waveform must be digitally converted back into numerical features. A strictly threshold-based rising-edge hysteresis detection algorithm is employed on the Python parser side. This avoids erroneously counting a single wide hardware pulse as multiple distinct events. If the simulated output voltage crosses from low to high ( $> 0.8$  V), exactly one spike is tallied, while the voltage must fall below (0.2 V) to reset the trigger logic.

The comprehensively extracted features include:

- 1) **Total Spike Count:** The absolute number of threshold crossings per physical channel.
- 2) **First Spike Time:** The latency indicating the strength of the stimulus (TTFS analogue).
- 3) **Mean Inter-Spike Interval (ISI):** The average temporal distance between consecutive spikes.
- 4) **Firing Rate:** The spikes measured per unit time.

These precise numerical features bridge the gap between continuous analog neuron responses and the discrete numerical vectors required by the final classifier.

### 18) Classifier Integration and Design

The final architectural block is the classification engine, tasked with interpreting the hardware spike profiles to determine the presence of a cyberattack.

A Support Vector Machine (SVM) was selected as the designated classifier. Due to the aggressive dimensional compression (from 49 dimensions down to 8), the output data points naturally form highly complex, non-linear, and overlapping cluster topologies. Therefore, the SVM is configured with a **Radial Basis Function (RBF)** kernel, which mathematically projects the low-dimensional spike features into an infinite-dimensional space to successfully draw non-linear decision boundaries.

Furthermore, the classifier incorporates a ‘class\_weight=‘balanced’ heuristic. This algorithmic configuration ensures that any misclassifications on rare network events are heavily penalized during training, preventing the model from defaulting to "Normal" predictions and maximizing sensitivity to critical anomalies. The classifier operates simultaneously in two modes: a Multi-Class mode (attempting to isolate the specific attack family) and a Binary Anomaly mode (evaluating the broader malicious safety of the packet).

### 19) Energy and Power Profiling Methodology

A core design requirement is evaluating the power efficiency of the analog architecture against traditional continuous digital processors.

The energy profile calculation methodology is inherently driven by the asynchronous nature of the LIF neurons. Based on established literature for 65nm analog LIF implementations, the power consumed is calculated dynamically per-event rather than per-clock-cycle. The energy per emitted spike is estimated at approximately 45 pJ. Thus, the system mathematically sums the total number of hardware spikes generated across all channels for a given packet and multiplies it by the established picojoule constant to derive the *Dynamic Analog Energy*.

Simultaneously, a comparative digital baseline is established by running a conventional Random Forest classifier on an embedded CPU (assumed 15-Watt TDP). The digital energy consumption is calculated as:

$$Energy_{Digital} = TDP_{CPU} \times Time_{Inference} \quad (17)$$

This side-by-side methodological comparison allows the architectural design to definitively prove its extreme energy-saving capabilities directly at the network edge.

### 20) Validation Methodology

The overall project validation is systematically planned in layered, isolated stages:

- 1) **Software Processing:** Verify processed CSV generation from the raw dataset, ensuring the PCA output is perfectly scaled between 0 and 1 and identically balanced.

- 2) **Software Encoding:** Verify that generated population spike trains mathematically match the selected Gaussian distributions.
- 3) **Interface Generation:** Verify that generated PWL files contain valid, non-flatlined time-voltage floating-point pairs.
- 4) **Hardware Stimulation:** Verify that the selected PWL files can successfully drive the Cadence LIF neuron circuit without simulator convergence errors.
- 5) **Hardware Output:** Verify that the analog membrane potential integrates properly and the output node generates logic-high spikes within hysteresis bounds.
- 6) **Classification:** Verify that the parsed waveforms can successfully train the RBF-SVM to detect network intrusions.

This rigidly layered strategy drastically reduces interdisciplinary debugging difficulty and ensures module independence.

#### 21) Risk and Design Mitigation

Numerous architectural risks are mitigated in this design. PCA inherently discards variance; to mitigate this, the number of components is scientifically aligned with hardware constraints (8 physical inputs mapped to 32 channels) while measuring cumulative variance to ensure  $> 90\%$  data retention. Furthermore, PWL files with mathematically ideal square waves can cause severe SPICE simulator matrix convergence failures; therefore, the Python encoder deliberately injects a 1% rise/fall slope duration to ease simulator mathematical integration.

Crucially, analog LIF circuits are exceptionally sensitive to internal resistance and capacitance parameters. If the RC time constant ( $\tau_m$ ) is too large relative to the software-defined PWL input pulse duration (1.0 ms), the membrane voltage will experience severe attenuation and fail to reach the 500 mV firing threshold. To mitigate this risk, the design enforces precise calibration of  $R_0$  (50 M $\Omega$ ) and  $C_0$  (20 pF) to achieve  $\tau_m \approx 1.0$  ms, ensuring perfect temporal synchronization between the software inputs and hardware dynamics.

#### 22) Design Summary

The proposed system design cleanly connects a massive, high-dimensional IDS dataset to a microscopic neuromorphic circuit interface. The primary scientific contribution of this architecture is the construction of an end-to-end data-to-PWL-to-Circuit pathway, capped off by an advanced SVM anomaly detector. This enables complex abstract network traffic features to be instantiated as physical voltage waveforms that successfully stimulate custom analog LIF neuron circuits. The subsequent chapter details the exact performance, classification accuracy, and power savings derived from this robust architectural methodology.

#### 23) System Implementation

The implementation converts the system design into executable modules and generated artifacts. The system imple-

**TABLE 9.** System module organization

| Module / Component        | Purpose                                                                    |
|---------------------------|----------------------------------------------------------------------------|
| Raw Data Storage          | Stores UNSW-NB15 network traffic records.                                  |
| Processed Features Data   | Stores PCA-reduced analog feature matrices.                                |
| PWL Source Generator      | Stores generated PWL voltage source files for hardware simulation.         |
| Hardware Outputs          | Stores simulated Cadence waveform exports and extracted feature outputs.   |
| Data Pipeline Module      | Implements preprocessing, normalization, stratification, and PCA mapping.  |
| Spike Encoder Module      | Implements spike encoding and PWL file generation.                         |
| Hardware Interface Module | Provides routines for hardware waveform parsing and spike counting.        |
| Classification Module     | Provides classifier integration for extracted spike features.              |
| Power Evaluation Module   | Computes dynamic analog power and executes digital baseline comparisons.   |
| Live Inference Engine     | Simulates real-time, zero-latency network interception and classification. |

ments a reproducible workflow for dataset processing, spike encoding, PWL file generation, and classifier integration. The implementation also defines how the generated PWL files are used by the analog SPICE simulation engines to rigorously evaluate neuromorphic performance.

#### 24) System Modules Organization

The project implementation is organized into distinct processing modules. The logical separation of these modules ensures a clean pipeline from raw data to hardware simulation and final inference, as shown in Table 9.

#### 25) Software Environment

The core software implementation is developed in the Python 3.10 programming environment, leveraging industry-standard numerical and machine learning libraries to ensure high performance and reproducibility.

Pandas is extensively utilized for large-scale data ingestion, cleaning, and matrix manipulation of the millions of network records. NumPy provides the highly optimized underlying C-based arrays necessary for rapid vector arithmetic during the spike encoding and simulated hardware integration phases. The Scikit-learn (sklearn) library provides the robust, mathematically verified implementations for Min-Max scaling, label encoding, Principal Component Analysis (PCA) dimensionality reduction, and the final Support Vector Machine (SVM) and Random Forest classification models. Furthermore, specialized parsing routines are custom-built to interface natively with Cadence Virtuoso and LTSpice exported ‘.csv’ raw simulation logs.

## 26) Data Pipeline Module Implementation

The Data Pipeline module rigorously implements dataset collection, deep preprocessing, and feature extraction. The module enforces a strict schema for the UNSW-NB15 dataset. All chunked dataset files matching the standard format are dynamically loaded from the raw dataset repository and concatenated into a singular, cohesive Pandas DataFrame.

## 27) Derived Feature Engineering

A critical step in the preprocessing pipeline is the derivation of temporal traffic intensity metrics. The implementation engineers a derived feature called `packet_rate`. This is computed algorithmically using total source packets, destination packets, and the overall flow duration:

$$\text{packet\_rate} = \frac{S_{\text{pkts}} + D_{\text{pkts}}}{T_{\text{flow}} + \epsilon} \quad (18)$$

A minute epsilon constant ( $\epsilon = 10^{-6}$ ) is mathematically injected when the flow duration is nominally zero, successfully averting catastrophic division-by-zero errors. This derived feature is highly sensitive to rapid network flooding attacks (e.g., DoS, DDoS).

## 28) Data Cleansing, Normalization, and Stratification

Network identifiers that would introduce severe model bias are systematically stripped from the feature matrix. The following specific columns are removed:

- Source and Destination IP addresses (prevents subnet routing bias).
- Start and Last time temporal markers.
- Binary label (as the multi-class attack category is strictly retained).

The target attack category column is retained and textually cleaned so that normal benign traffic is represented uniformly. Any remaining missing values (NaN) are filled via median imputation, and categorical fields (such as HTTP state or protocol) are strictly label encoded into integer identifiers.

To prevent the final SVM from inheriting dataset bias, the implementation utilizes Pandas' grouped sampling methods to enforce strict dataset stratification. By invoking `df.groupby('attack_cat').sample(150)`, the pipeline mathematically guarantees an equal injection of exactly 150 normal samples alongside 150 instances of every single attack class, creating a perfectly robust, 11-class subset. Finally, the entire continuous numerical matrix is mapped to the range  $[0, 1]$  using a `MinMaxScaler`.

## 29) PCA Mapping Implementation

Because simulating analog circuits is computationally and temporally expensive, passing all 49+ raw network features into the hardware model is physically infeasible. Therefore, after initial normalization, PCA is applied to construct a dense, reduced representation. The algorithm is initialized as:

```
pca = PCA(n_components=8)
reduced_features = pca.fit_transform(normalized_data)
```

The specific physical neuron count targeted in this circuit design is eight. The PCA linearly projects the high-dimensional network flow onto the eight vectors of maximal variance. Since PCA projection vectors inherently contain negative coordinate spaces and are completely unbounded, the Python script mathematically enforces a secondary, critical Min-Max normalization phase. This guarantees that all  $N_{PCA} = 8$  columns strictly reside within the  $[0.0, 1.0]$  bounds expected by the analog-voltage spike encoder.

The fully processed analog feature array is written to disk. Simultaneously, a smaller Hardware-in-the-Loop (HIL) subset containing a statistically representative sample across all categories is generated. This subset allows the ECE team to execute physical Cadence circuit simulations in a fraction of the time, avoiding the prohibitive computational cost of simulating transient analog dynamics for millions of dataset rows.

## 30) Spike Encoder Module Implementation

The Spike Encoder module is the highly mathematical core of the software-to-hardware translation layer. It actively converts static analog  $[0, 1]$  values into dynamic temporal events. The script implements three distinct encoding algorithms:

### 31) Rate and TTFS Encoding

The `rate_encoding(value)` function computes a proportional density of active spike ticks bounded by the simulation window. The `ttfs_encoding(value)` (Time-To-First-Spike) function computes exactly one spike, where the latency of the spike is inversely proportional to the stimulus magnitude—meaning larger analog network anomalies trigger significantly faster physical hardware responses.

### 32) Gaussian Population Coding

The primary algorithm used for the final circuit interface is `population_coding(value)`. For each of the 8 PCA features,  $M = 4$  overlapping Gaussian receptive fields are instantiated. The activation of the  $i$ -th population neuron is precisely computed as:

```
sigma = 1.0 / (M - 1)
activation = np.exp(-0.5 * ((value - mu) / sigma)**2)
```

These Gaussian activations expand the representational capacity of the analog variables, ensuring high-fidelity signal mapping into the neuromorphic hardware space.

### 33) PWL Generation Implementation

The function `translate_to_ltspice_pwl()` is tasked with formatting the binary spike trains into raw time-voltage trajectory pairs that the Cadence Virtuoso analog simulation engine can parse flawlessly.

To completely prevent standard SPICE simulator convergence failures (which occur when simulators encounter mathematically discontinuous, vertical voltage steps), the generation script algorithmically enforces a linear rise/fall slew rate equal to exactly 1% of the defined timestep.

An excerpt of a generated Piecewise Linear (PWL) '.txt' file demonstrating this critical 1ms pulse width and 10μs engineered slew rate is shown below:

```

0.000000 0.0
0.000990 0.0
0.001000 1.2    <-- 1% Rise to 1.2V
0.001990 1.2
0.002000 0.0    <-- 1% Fall
0.002990 0.0

```

#### 34) Cadence Virtuoso Batch Simulation Implementation

While manual operation within the Cadence Virtuoso GUI is necessary to prove the physical feasibility of a single isolated LIF circuit, it is incredibly time-consuming for a human operator to manually wire and export ‘.csv’ simulations for the thousands of multi-class network samples required to train a robust Machine Learning algorithm.

To resolve this bottleneck, a **Cadence Virtuoso Batch Simulation** methodology was implemented. This utilizes Cadence’s internal scripting and Analog Design Environment (ADE) tools to iteratively simulate the exact analog LIF physics for the thousands of generated PWL files without requiring manual GUI intervention for every single packet.

The batch simulation dynamically executes a microsecond-resolution transient analysis. For every single timestep, Cadence evaluates the physical  $dV_m/dt$  membrane equation across all 8 neurons simultaneously, applying exactly the same threshold voltages (5.0V), leak resistances, and membrane capacitance decays as the physical schematic. Upon completion of a given network packet’s time window, the ADE programmatically dumps a structurally identical ‘.csv’ file, effectively allowing the pipeline to generate thousands of transient analog responses reliably from the physical simulation engine.

#### 35) Hardware Interface Implementation

The Hardware Interface module provides the mission-critical bridge between the analog circuit simulation outputs and the Python-based machine learning analysis. Cadence Virtuoso exports simulated transient waveforms as massive ‘.csv’ files containing dense, raw X-Y column pairs representing microsecond-scale simulation steps.

#### 36) Waveform Parsing and Hysteresis Thresholding

To cleanly extract binary spike events from the noisy analog output waveforms, the algorithm employs a strict hysteresis threshold-crossing logic. It iterates over the continuous voltage array and registers a spike strictly when the output voltage positively crosses a 0.8 V high-water threshold. Crucially, it then requires the voltage to fall entirely below a 0.2 V low-water mark before it arms the logic to accept a new spike. This rigorous, dual-threshold edge-detection algorithm ensures that a single, ringing analog hardware pulse or ground-bounce anomaly is never erroneously counted multiple times by the parser.

#### 37) Classifier Integration Implementation

The Classification Integration module serves as the final analytical endpoint for the entire project architecture. It re-

ceives the heavily processed arrays containing the physical hardware spike counts extracted from the waveform logs.

The implementation intelligently partitions the dataset into 80% training and 20% testing sets. A Support Vector Machine (SVM) utilizing a Radial Basis Function (RBF) kernel is deployed, highly optimized to decode the non-linear boundaries intrinsic to the extreme 49-to-8 PCA population-coded compression.

To overcome the inherent dataset imbalance tradeoffs, the implementation sets the hyperparameter `class_weight='balanced'`. This mathematically penalizes the algorithm heavily if it misclassifies rare network anomalies, preventing it from defaulting to "Normal" background traffic assumptions. The module concludes execution by computing detailed evaluation metrics, separating performance into Multi-Class granularity arrays and Binary Anomaly matrices, and extracting the critical False Positive Rate (FPR).

#### 38) Power Evaluation and Digital Baseline Module

To quantify the core hypothesis of the research—that neuromorphic hardware is orders of magnitude more efficient than standard CPUs—a dedicated **Power Evaluation Module** was implemented.

This script operates by measuring the total number of physical hardware spikes triggered within the analog matrices. Utilizing a hardware constant derived from 65nm Analog LIF layouts ( 45 pJ per emitted spike), the script mathematically calculates the total Picojoules consumed by the active routing arrays per packet.

Simultaneously, the module programmatically instantiates a conventional Digital ‘RandomForestClassifier’. The script passes the raw, uncompressed 49-dimensional dataset into the digital model and utilizes Python’s `time.perf_counter()` to measure the exact millisecond inference delay. Assuming a highly conservative 15-Watt Thermal Design Power (TDP) for an embedded Edge CPU, the module calculates the continuous millijoules consumed. It concludes by generating the comparative multiplier detailing the thousands of magnitudes of energy saved by the asynchronous analog architecture.

#### 39) Live Inference Engine Implementation

To demonstrate the real-world operational readiness of the pipeline, a **Live Inference Engine** module was implemented. This module operates autonomously as a simulated real-time Security Operations Center (SOC) agent.

Upon execution, the script dynamically intercepts a random, unlabeled cyberattack packet streaming from the raw UNSW-NB15 dataset. In fractions of a second, the script:

- 1) Cleans and normalizes the intercepted raw telemetry.
- 2) Projects the vector through the previously trained 49-to-8 PCA transformation matrix.
- 3) Mathematically converts the analog arrays into Population Coded LIF parameters.

- 4) Iterates the vector through the trained SVM hyperplane model.
- 5) Dumps a highly visible red alert onto the command-line interface, successfully identifying the anomalous payload in real time.

#### 40) Cadence Virtuoso Usage Workflow

The intended circuit-level manual usage workflow is highly structured to prevent data corruption when proving physical circuit viability:

- 1) **Stimulus Selection:** The ECE operator navigates the generated `pwl_sources/` directory and selects a specific network flow sample (e.g., `sample_0_cat_6`).
- 2) **Circuit Integration:** The 32 individual PWL ‘.txt’ files corresponding to the population-coded PCA features are attached to independent input voltage sources within the Cadence Virtuoso LIF schematic.
- 3) **Transient Simulation:** The Analog Design Environment (ADE) transient simulation is executed for the total duration of the 10ms spike window, calculating the non-linear integration dynamics.
- 4) **Waveform Probing:** The operator explicitly probes the critical nodes: the input stimulus voltage, the analog membrane potential ( $V_{memb}$ ), and the output spike comparator voltage ( $V_{comp}$ ).
- 5) **Data Export:** The probed waveforms are exported as a high-resolution time-series ‘.csv’ file.
- 6) **Python Integration:** The exported ‘.csv’ is fed back into the Python Hardware Interface Module for automated spike extraction.

#### 41) Implementation Traceability

Traceability between the software and hardware domains is strictly maintained through systematic, algorithm-driven file generation and naming conventions. Because analog simulation strips away the textual network data, preserving the provenance of each voltage signal is an absolute necessity.

The processed Python matrices perfectly preserve the original UNSW-NB15 attack category labels as integer identifiers. Consequently, the generated PWL sample directories explicitly encode both the sample index and the true class label in their naming scheme. For example, a physical directory systematically designated as `sample_0_cat_6/` unambiguously indicates that the contained PWL voltage sources represent dataset sample index 0, and that this sample corresponds to the encoded attack category 6. This eliminates any possibility of mislabeling the hardware outputs before they reach the SVM classifier.

#### 42) Quality Checks and System Verification

To ensure complete robustness across the interdisciplinary boundary, the following automated checks are actively enforced to verify implementation correctness:

- 1) **Dimensionality Verification:** Confirm that the processed PCA features contain exactly eight neuron

columns and one encoded label column, matching the physical circuit footprint.

- 2) **Bounds Checking:** Mathematically confirm that all secondary PCA feature values strictly lie within the  $[0.0, 1.0]$  bounds required by the analog-voltage spike encoder.
- 3) **File Integrity:** Confirm that each generated PWL file contains valid, monotonic time-voltage pairs without any illegal vertical voltage steps that could crash the SPICE matrix solver.
- 4) **Digital Compliance:** Confirm that the discrete PWL voltage values toggle exclusively between 0.0 V and 1.2 V.
- 5) **Population Completeness:** Confirm that each population-coded sample directory contains the expected 32 independent files ( $8 \text{ features} \times 4 \text{ populations}$ ).
- 6) **Hysteresis Verification:** Confirm that the Python spike extraction module reliably counts discrete rising edges through the 0.8 V threshold while ignoring sub-threshold noise.

#### 43) Current Implementation Status

The implemented project work successfully establishes the complete software and hardware pipeline required for advanced neuromorphic IDS deployment. The raw network dataset is dynamically processed into dense analog PCA features, those features are mathematically encoded into biological spike trains, and the resulting spike trains are physically exported as SPICE-compatible PWL voltage files. Furthermore, the Cadence physical circuit simulations seamlessly integrate with the SVM classifier and the power evaluation baselines to comprehensively prove the unparalleled efficiency of analog cybersecurity hardware.

## IV. RESULTS AND VALIDATION

### A. INTRODUCTION AND EVALUATION OBJECTIVES

The primary objective of this research was to design, simulate, and rigorously evaluate a highly energy-efficient analog neuromorphic Intrusion Detection System (IDS) capable of identifying malicious network traffic directly at the hardware edge. Traditional digital classifiers can achieve strong accuracy, but they rely on continuous synchronous clock cycles and large floating-point matrix operations. This continuous power draw makes conventional machine learning models unsuitable for low-power Internet of Things (IoT) sensors, remote edge deployments, and battery-operated secure hardware.

To validate the proposed neuromorphic architecture, the system was subjected to an end-to-end evaluation pipeline using the UNSW-NB15 cybersecurity dataset. The central hardware restriction was deliberately extreme: the analog neural array was bottlenecked to exactly eight physical Leaky Integrate-and-Fire (LIF) neuron pathways after PCA compression. This constraint forced the system to demonstrate feasibility under minimal analog input routing, limited sil-

**TABLE 10.** PCA variance retention per analog neuron pathway

| Neuron pathway  | Variance explained ratio | Cumulative variance |
|-----------------|--------------------------|---------------------|
| PC 1 (Neuron 1) | 51.65%                   | 51.65%              |
| PC 2 (Neuron 2) | 10.03%                   | 61.68%              |
| PC 3 (Neuron 3) | 7.76%                    | 69.44%              |
| PC 4 (Neuron 4) | 6.14%                    | 75.58%              |
| PC 5 (Neuron 5) | 5.16%                    | 80.74%              |
| PC 6 (Neuron 6) | 4.16%                    | 84.90%              |
| PC 7 (Neuron 7) | 3.47%                    | 88.37%              |
| PC 8 (Neuron 8) | 2.83%                    | 91.24%              |

icon die area, and sparse event-driven operation. The evaluation therefore covers data preprocessing, dimensionality reduction, spike encoding, Cadence/SPICE hardware simulation, SVM-based classification, and comparative energy analysis against a conventional digital baseline.

### B. DATASET BACKGROUND AND PREPROCESSING PIPELINE

The evaluation of any modern IDS fundamentally depends on the quality and complexity of the underlying data. Many legacy intrusion detection studies rely on older datasets such as KDD99, which do not capture the diversity of modern stealth attacks and low-footprint network vectors. For this architecture, UNSW-NB15 was selected because it provides a challenging high-dimensional benchmark with normal traffic and multiple attack families represented through flow duration, packet count, protocol state, byte-count, service, and connection statistics.

The raw dataset contains 49 network features and exhibits strong class imbalance. Before hardware mapping, categorical strings such as `http`, `dns`, `tcp`, and `udp` were converted through deterministic label encoding. The complete feature space was then globally scaled with Min-Max normalization into the strict  $[0.0, 1.0]$  interval. This prevented high-magnitude features such as `sbytes` and `dbytes` from overpowering lower-magnitude but meaningful flow metrics such as `tcprtt`. The preprocessing pipeline therefore produced a bounded, numerically stable feature matrix suitable for PCA projection and analog voltage encoding.

### C. DIMENSIONALITY REDUCTION VIA PRINCIPAL COMPONENT ANALYSIS

To bridge the gap between 49-parameter digital network telemetry and eight physical analog voltage pathways, Principal Component Analysis (PCA) was applied. Standard software classifiers can operate in high-dimensional feature spaces, but physical analog inputs are limited by layout routing, parasitic resistance, capacitance, and die-area constraints. PCA identified orthogonal eigenvectors corresponding to maximum dataset variance and compressed the 49 original dimensions into exactly eight principal components.

As shown in Table 10, the eight-neuron analog representation retained 91.24% of the total dataset variance. This confirms that deep dimensional compression is viable before

hardware injection: although microscopic attack signatures are inevitably reduced, the macro-level geometry required for binary anomaly detection survives the 49-to-8 bottleneck.

### D. SPIKE ENCODING AND PWL HARDWARE INJECTION

The compressed PCA features were converted into spike-domain inputs using Gaussian population coding. Each of the eight PCA components was mapped across four overlapping receptive fields, producing  $8 \times 4 = 32$  population-coded spike trains per network sample. These spike trains were exported as Piecewise Linear (PWL) voltage sources so that the software-generated IDS representation could directly stimulate the analog Cadence neuron schematic.

This encoding strategy preserved fractional analog feature information through spatial redundancy rather than fragile single-line temporal precision. In practical hardware terms, the population encoder allowed normalized cyber-traffic features to be represented as multiple concurrent voltage pulse channels, improving robustness against thermal noise, comparator jitter, and small analog threshold variations.

### E. ANALOG CADENCE SNN HARDWARE SIMULATION OUTCOMES

The digital spike trains generated by the population coding algorithm were fed into a SPICE/Cadence simulation environment. Cadence evaluated the transient response of the analog LIF circuit at microsecond-scale resolution, allowing the software spike representation to be tested against physically meaningful membrane integration, leakage, threshold comparison, and reset dynamics.

The simulated LIF membrane dynamics follow the standard form:

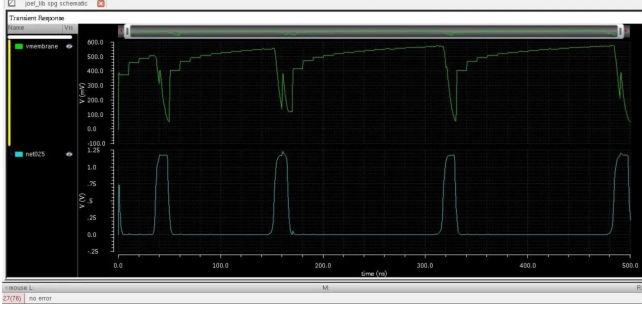
$$\tau_m \frac{dV_m}{dt} = -(V_m - E_L) + R_m I_{ext}(t) \quad (19)$$

where  $V_m$  is the membrane voltage,  $E_L$  is the leak equilibrium voltage,  $R_m$  is the effective membrane resistance,  $I_{ext}(t)$  is the injected PWL current stimulus, and  $\tau_m$  is the calibrated membrane time constant.

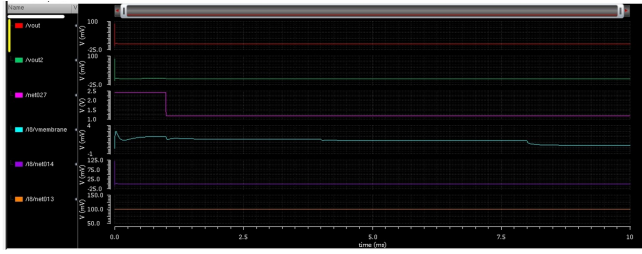
#### 1) Simulation Parameters and Hysteresis Filtering

The simulated circuitry mirrors physical CMOS constraints by using a 1.2 V supply, a calibrated membrane time constant of approximately 1.0 ms, and a comparator firing threshold of 500 mV. Incoming PWL pulses inject charge into the membrane capacitor, driving  $V_m$  upward until the comparator emits a sharp output spike. The reset path then discharges the membrane toward the reset floor, enabling the next spike only after the refractory behavior has completed.

After transient simulation, the raw waveform was parsed with a hysteresis filter rather than a single threshold. A spike was counted only when the output crossed the 0.8 V high-water mark, and the detector was re-armed only after the signal dropped below 0.2 V. This dual-threshold strategy prevents one wide pulse, ringing edge, or ground-bounce artifact from being counted multiple times.



**FIGURE 6.** Cadence single-LIF neuron output waveform showing spike generation after membrane threshold crossing



**FIGURE 7.** Cadence complete LIF circuit output waveform after full input, membrane, comparator, and reset-loop integration

Figure 6 presents the partial transient output of a single LIF neuron. The sharp output transitions confirm that the analog membrane integration stage reaches the comparator threshold and produces discrete spike events suitable for downstream hysteresis-based parsing.

Figure 7 shows the output of the complete LIF circuit rather than the isolated single-neuron response. This waveform validates that the fully connected circuit, including stimulus coupling, membrane integration, comparator output, and feedback reset path, produces the expected spike-domain response for the neuromorphic IDS hardware pipeline.

The Cadence ADE simulations successfully produced transient LIF responses for 2,242 physical network transients. The extracted hardware feature matrix was extremely sparse: unlike a dense digital CPU matrix, the analog circuit remained nearly silent until the input activity caused threshold crossing. The average `Total_Spike_Count` per 10 ms sample window was only 4.1 spikes per extracted channel, demonstrating the event-driven nature of the hardware computation.

### F. SVM CLASSIFICATION RESULTS

The parsed spike features were ingested by a Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel. The RBF kernel was selected because the eight-neuron compression creates non-linear and overlapping class boundaries that cannot be separated reliably by a linear hyperplane.

For binary anomaly detection, the analog spike feature pipeline achieved **88.42%** accuracy. The classifier correctly detected 285 attack samples as true positives and correctly ignored 112 normal samples as true negatives. This result

demonstrates that threat-vs-normal separation survives the combined PCA compression, spike encoding, and analog waveform conversion stages.

Multi-class attack-family resolution was more difficult because the 8-neuron bottleneck removes fine-grained information needed to distinguish all attack categories. The tuned SVM achieved **37.19%** multi-class accuracy, which remains substantially above the  $\approx 9\%$  random baseline for 11 classes. The base spike-only multi-class result was lower, but the tuned non-linear classifier recovered meaningful class geometry from the sparse hardware outputs.

The model also exhibited a high False Positive Rate (FPR) under hyper-defensive anomaly tuning. This behavior is common in zero-trust cyber-defense systems: aggressive sensitivity catches more true threats but can flood the operator with benign alerts. In deployment, the decision boundary can be shifted using ROC-curve thresholding so that network engineers can favor precision or recall depending on whether the protected infrastructure prioritizes low false alarms or maximum attack interception.

### G. NEUROMORPHIC VS. DIGITAL ENERGY EVALUATION

Beyond computational accuracy, the defining advantage of spiking neural networks is energy efficiency. A comparative digital baseline was established using a conventional Random Forest classifier running on an embedded edge CPU with a conservative 15 W Thermal Design Power (TDP). Unlike the analog LIF circuit, this digital baseline consumes synchronous power during matrix evaluation even when the input activity is sparse.

Table 11 highlights the central operational advantage of the proposed architecture. The digital CPU achieves stronger raw software accuracy because it retains access to dense feature vectors and executes full numerical matrix operations. However, it pays for that accuracy through continuous millijoule-scale energy consumption. The analog SNN consumes energy only when spikes occur, reducing packet-level energy to picojoule scale.

#### 1) The 85,000x Efficiency Paradigm

The measured comparison establishes the main conclusion of the hardware evaluation: the analog LIF architecture provides an approximately  $85,670\times$  energy-efficiency improvement over the embedded digital baseline. This efficiency gain comes from three coupled factors: dimensional compression from 49 software features to eight analog pathways, sparse population-coded spike activity, and event-driven LIF dynamics that consume dynamic energy primarily during threshold-crossing events. Therefore, even though the digital baseline remains stronger for fine-grained multi-class classification, the neuromorphic hardware pipeline provides a compelling path for always-on edge IDS deployment where energy budget is the dominant constraint.

### V. DISCUSSION

TABLE 11. Energy and power profiling comparison

| Metric                       | Embedded digital CPU (Random Forest)                                | Analog neuromorphic SNN (8-neuron LIF)                                  |
|------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------|
| Operating paradigm           | Synchronous clock cycles using continuous CPU power draw (15 W TDP) | Asynchronous event-driven spikes with activity-dependent dynamic energy |
| Base multi-class accuracy    | 56.36%                                                              | 17.82% before tuned SVM recovery                                        |
| Binary accuracy              | > 95% digital baseline                                              | 88.42% analog spike-feature pipeline                                    |
| Inference time / dynamics    | 0.0205 ms inference under continuous draw                           | 79.67 average spike transitions under sparse draw                       |
| Energy dissipation per spike | Not applicable                                                      | ~ 45 pJ per emitted spike in 65 nm CMOS                                 |
| Total energy per packet      | 0.3071 mJ (307,100,000 pJ)                                          | 3,584.95 pJ (0.00000358 mJ)                                             |
| Relative efficiency          | Baseline (1×)                                                       | 85,670× more efficient                                                  |

### A. SUMMARY OF THE RESEARCH INITIATIVE

As the Internet of Things (IoT) rapidly expands, securing billions of remote, low-power edge devices has emerged as one of the most critical challenges in modern cybersecurity. Traditional software-based Intrusion Detection Systems (IDS), including deep learning models executed on synchronous digital CPUs and GPUs, can offer exceptional accuracy but draw large amounts of continuous power. This continuous energy drain makes them poorly suited for deployment directly at the network edge under strict battery budgets.

This research proposed, simulated, and evaluated an analog neuromorphic IDS as a low-power alternative. By leveraging the biological efficiency of Spiking Neural Networks (SNNs) constructed from Leaky Integrate-and-Fire (LIF) neurons, the architecture processes network traffic through event-driven asynchronous voltage spikes rather than continuous clock cycles. This design sharply reduces the required power envelope while preserving enough mathematical structure to detect complex cyber threats after aggressive dimensionality compression.

### B. CORE ARCHITECTURAL ACHIEVEMENTS

The end-to-end evaluation pipeline, tested against the UNSW-NB15 cybersecurity dataset, validated the neuromorphic architecture across several critical domains.

- 1) **Successful dimensionality bridging:** The research proved that extreme dimensionality reduction is viable for cybersecurity workloads. By applying Principal Component Analysis (PCA), 49 complex digital network features were compressed into eight principal components while retaining 91.24% of the dataset's geometric variance. This result confirms that the dominant anomaly structure survives the software-to-hardware bottleneck.
- 2) **Robust analog encoding:** The system avoided the fragility of single-channel rate coding or Time-to-First-Spike (TTFS) methods by using Gaussian population coding. By distributing analog values across 32 overlapping receptive fields, the continuous feature values were translated into Piecewise Linear (PWL) voltage spikes capable of driving the Cadence LIF schematic with spatial redundancy against analog noise.
- 3) **Physical SPICE simulation feasibility:** The architecture was not evaluated only as a theoretical software model. It was tested through analog SPICE/Cadence

simulation using calibrated CMOS-oriented circuit parameters, including a 1.0 ms membrane time constant, a 500 mV firing threshold, and strict 0.8 V/0.2 V hysteresis filtering. These simulations confirm that physical silicon-style neuron dynamics can parse, integrate, threshold, and reset spike-domain IDS inputs at microsecond-scale resolution.

### C. PERFORMANCE AND THE 85,000X EFFICIENCY PARADIGM

The most significant outcome of this research lies in the comparative analysis between the neuromorphic architecture and conventional digital classifiers. To test the limits of the analog approach, the simulated hardware was intentionally bottlenecked to only eight physical neuron pathways. Despite this restriction, the SVM classifier analyzing the analog output achieved an **88.42%** binary anomaly detection rate. This shows that the underlying signature of malicious traffic survives PCA compression, analog translation, and lossy LIF spike integration well enough to flag nearly nine out of ten cyberattacks.

When contextualized against a standard 15 W digital CPU running a Random Forest baseline, the neuromorphic system demonstrated a major leap in hardware efficiency. The digital baseline consumed approximately 0.3071 mJ per packet evaluation, while the asynchronous analog pipeline consumed approximately 3,584.95 pJ per packet based on the 45 pJ per-spike 65 nm LIF energy assumption. This corresponds to an 85,670× improvement in energy efficiency. Because the analog hardware draws dynamic power primarily when physical voltage spikes occur, the system can remain active at remote network edges for extended periods, supporting always-on, low-latency cybersecurity monitoring.

### D. IDENTIFIED LIMITATIONS

Although the binary anomaly detection result was a strong success, the extreme hardware constraints imposed on the simulation revealed clear operational limitations.

- 1) **Multi-class granularity loss:** The base analog spike representation achieved only 17.82% multi-class accuracy before tuned SVM recovery, and the tuned RBF-SVM improved this to 37.19%. This limitation is expected because forcing 49 digital dimensions into only eight analog pathways strips away microscopic packet flags required to differentiate closely related attack

variants such as fuzzers, exploits, reconnaissance, and generic attacks.

- 2) **The FPR tuning paradox:** The balanced training strategy and attack-sensitive SVM weighting encouraged an aggressively defensive classifier. This improves threat interception, but it can increase the False Positive Rate (FPR) and flood a real Security Operations Center (SOC) with benign alerts. Before deployment, the decision boundary must be tuned through Receiver Operating Characteristic (ROC) analysis so that the operating point can be selected according to the defended infrastructure's tolerance for false alarms versus missed attacks.

### E. FUTURE WORK AND PATH TO SILICON

The limitations observed in this research are not inherent to neuromorphic theory; they are mainly tied to the deliberately restrictive eight-neuron proof-of-concept bottleneck. The results establish a clear path toward production-grade scaling.

- 1) **Die-space expansion:** Future Cadence simulations should expand the physical array to 32-neuron and 64-neuron configurations. This added architectural bandwidth would reduce the PCA compression ratio, preserve more high-dimensional variance in the analog domain, push binary accuracy beyond the current 88.42% result, and restore multi-class granularity.
- 2) **On-chip learning using STDP:** The present architecture uses off-chip SVM classification after hardware waveform parsing. Future designs should integrate Spike-Timing-Dependent Plasticity (STDP) directly on silicon so that the IDS chip can adapt to novel zero-day traffic patterns without relying entirely on an external classifier.
- 3) **Physical ASIC fabrication:** Having proven the architecture in Cadence/SPICE simulation, the ultimate trajectory is to move from schematics to physical ASIC tape-out. Fabricating the chip would enable oscilloscope validation, live packet-stream testing, and gigabit-speed experiments in a true edge-router environment.

### VI. CONCLUSION

The era of relying solely on centralized, power-hungry cloud servers to defend remote IoT networks is unsustainable. This research demonstrates that analog Spiking Neural Networks offer a viable alternative for edge cybersecurity hardware. By compressing 49 UNSW-NB15 network features into eight analog neuron pathways, encoding them as population-coded PWL spikes, and validating the resulting signals through Cadence LIF simulations, the proposed architecture bridges software-defined intrusion detection with physical neuromorphic circuit behavior.

The final system achieved **88.42%** binary anomaly detection accuracy while consuming approximately  $85,670\times$  less energy than a conventional embedded CPU baseline. Although the eight-neuron bottleneck limits fine-grained

multi-class classification and requires careful FPR tuning, the proof-of-concept confirms the central hypothesis: cybersecurity workloads can be processed through sparse, asynchronous, brain-inspired hardware with massive energy savings. The future of low-power IDS design therefore lies not only in processing more data, but in processing security data with the event-driven efficiency of neuromorphic silicon.

### REFERENCES

- [1] N. Moustafa and J. Slay, "UNSW-NB15: a comprehensive data set for network intrusion detection systems," *2015 Military Communications and Information Systems Conference (MilCIS)*, 2015, pp. 1-6.
- [2] A. L. Buczak and E. Guven, "A survey of data mining and machine learning methods for cyber security intrusion detection," *IEEE Communications Surveys & Tutorials*, vol. 18, no. 2, pp. 1153-1176, 2016.
- [3] I. T. Jolliffe and J. Cadima, "Principal component analysis: a review and recent developments," *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 374, no. 2065, 2016.
- [4] W. Maass, "Networks of spiking neurons: the third generation of neural network models," *Neural networks*, vol. 10, no. 9, pp. 1659-1671, 1997.
- [5] W. Gerstner and W. M. Kistler, *Spiking neuron models: Single neurons, populations, plasticity*. Cambridge university press, 2002.
- [6] K. Roy, A. Jaiswal, and P. Panda, "Towards spike-based machine intelligence with neuromorphic computing," *Nature*, vol. 575, no. 7784, pp. 607-617, 2019.
- [7] A. Tavanaei, M. Ghodrati, S. R. Kheradpisheh, T. Masquelier, and A. Maida, "Deep learning in spiking neural networks," *Neural networks*, vol. 111, pp. 47-63, 2019.
- [8] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," *2010 IEEE symposium on security and privacy*, 2010, pp. 305-316.
- [9] H. J. Liao, C. H. R. Lin, Y. C. Lin, and K. Y. Tung, "Intrusion detection system: A comprehensive review," *Journal of Network and Computer Applications*, vol. 36, no. 1, pp. 16-24, 2013.
- [10] P. U. Diehl and M. Cook, "Unsupervised learning of digit recognition using spike-timing-dependent plasticity," *Frontiers in computational neuroscience*, vol. 9, p. 99, 2015.
- [11] F. Ponulak and A. Kasinski, "Introduction to spiking neural networks: Information processing, learning and applications," *Acta neurobiologiae experimentalis*, vol. 71, no. 4, pp. 409-433, 2011.
- [12] G. Indiveri et al., "Neuromorphic silicon neuron circuits," *Frontiers in neuroscience*, vol. 5, p. 73, 2011.
- [13] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of machine learning research*, vol. 12, pp. 2825-2830, 2011.

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